

Learn Java Syntax

- Websites:
<https://www.tpointtech.com/java-tutorial>
<https://www.tutorialspoint.com/java/index.htm>
<https://www.w3schools.com/java/> (my fav)
- Youtube:
https://www.youtube.com/playlist?list=PLsyebzWxl7pe_liTfNyr55kwJPWbgxB5

Learn OOP (Theory -> Code)

- <https://www.youtube.com/playlist?list=PLsyebzWxl7oa1WO9n4cP3OY9nOtUcZlg>
- <https://www.youtube.com/watch?v=XPCG24IEGTo> (has questions as well)

Learn Collections

- https://www.youtube.com/watch?v=VE_AAUxTUCY

Learn Advanced Java Topics

- Concurrency + Multi Threading :
<https://www.youtube.com/watch?v=WldMTtUWqTg>
- Exception Handling:
<https://www.youtube.com/watch?v=IZu5rZTN7PI>
<https://www.youtube.com/watch?v=zVRvPXbhKEo>
- Lambda:
https://youtu.be/0ada8fAMpVs?si=8XP0V7-i2_ftbs8w

OOP Project (Do and Upload code to Github)

Java OOP Mini Project — *Battle Arena Game*

Problem Statement:

Design and implement a Battle Arena game in Java where multiple types of characters (like warriors, mages, archers, etc.) can fight each other using different attacks, weapons, and abilities.

Requirements:

1. Create an abstract class `Character` with common attributes (e.g., name, health, attack power, defense) and abstract methods like `attack()` and `takeDamage()`.

2. Create at least three subclasses of `Character` (for example, `Warrior`, `Mage`, `Archer`), each with its own special behavior.
3. Create one or more interfaces (e.g., `Weapon`, `SpecialAbility`) and make your classes implement them appropriately.
4. Use collections (`ArrayList`, `HashMap`, etc.) wherever suitable — for example, to store characters, weapons, or battle results.
5. Implement a `BattleArena` class that manages the fight between characters (turn-based or simulated automatically).
6. The game should display round-by-round results and announce the winner at the end.
7. Use clean OOP principles — abstraction, inheritance, polymorphism, and encapsulation.

Bonus (Optional):

- Add a scoreboard system using a `HashMap`.
- Allow random weapon assignment or special abilities.
- Let the user choose a character to play as, and simulate the enemy automatically

DSA

- <https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2/>
- Leetcode Contests

Frontend (For FullStack)

- Learn HTML, CSS, JS + React:
https://www.youtube.com/watch?v=6mbwJ2xhgzM&list=PLu0W_9lI9agiCUZYRs_vtGTXdxkzPyItg
- https://www.youtube.com/playlist?list=PLu0W_9lI9agx66oZnT6lyhcMibUMNMdt

DBMS + SQL

- <https://www.geeksforgeeks.org/dbms/dbms/>
- <https://www.w3schools.com/sql/>

JDBC + Other Concepts

- https://www.youtube.com/playlist?list=PLIhM4lkb2sEjVsbbZ_kiixY5CcR84lQUg
- <https://www.youtube.com/watch?v=OuBUUkQfBYM&t=6403s>
- <https://www.youtube.com/playlist?list=PLsyebzWxl7rU7Jz3zDRpqB-EODzBbHOI>

SpringBoot

- https://www.youtube.com/playlist?list=PLsyebzWxl7oA8QOIMtQsRT_I7Rx2hoX4
- https://www.youtube.com/playlist?list=PL0zysOfIRCelmjxj-g4jLr3WKraSU_e8q
- <https://www.youtube.com/watch?v=sdDDuQuX2cg> (RestAPI Project)

Full Stack Project

- <https://www.youtube.com/watch?v=9i1gQ7w2V24>

Microservices, Advanced Backend

- https://www.youtube.com/playlist?list=PL7CBVLpg0zqeJTCCuU99RW1TMn8_li15n
- <https://www.youtube.com/watch?v=fmX84zu-5gs>
- **MORE TO BE ADDED SOON..**

